

Contents

1	Sketches of the System: In Paper	3
2	Figure Family A: Appendix	5
3	Figure Family B	7
4	Figure Family C	8
5	Figure Family F: In Paper	9
6	Figure Family G: In Paper	11
7	Figure Family H	13
8	Figure Family I: Not for paper	17
9	Figure J	18
10	Figure K: In paper	18
11	Figure Family D-OLD: Replaced by F	19
12	Figure Family E-OLD: Replaced	23

Standard Parameters of the system:

- Time increment $dt = 10^{-5}s$
- Mass $M = 0.004kg$
- Radius $R = 0.005m$
- Grav. acc. $g = 9.80665m/s^2$
- Outer Magnetic Field $B_x = B_y = 0, B_z = 5 \cdot 10^{-5}T$
- Initial values $x_0 = y_0 = 0$
- Rolling Resistance coef. $\nu = 0.002$
- Mass density of air $\rho = 1.2kgm^3$
- Drag coefficient of sphere $c = 0.47$

1 Sketches of the System: In Paper

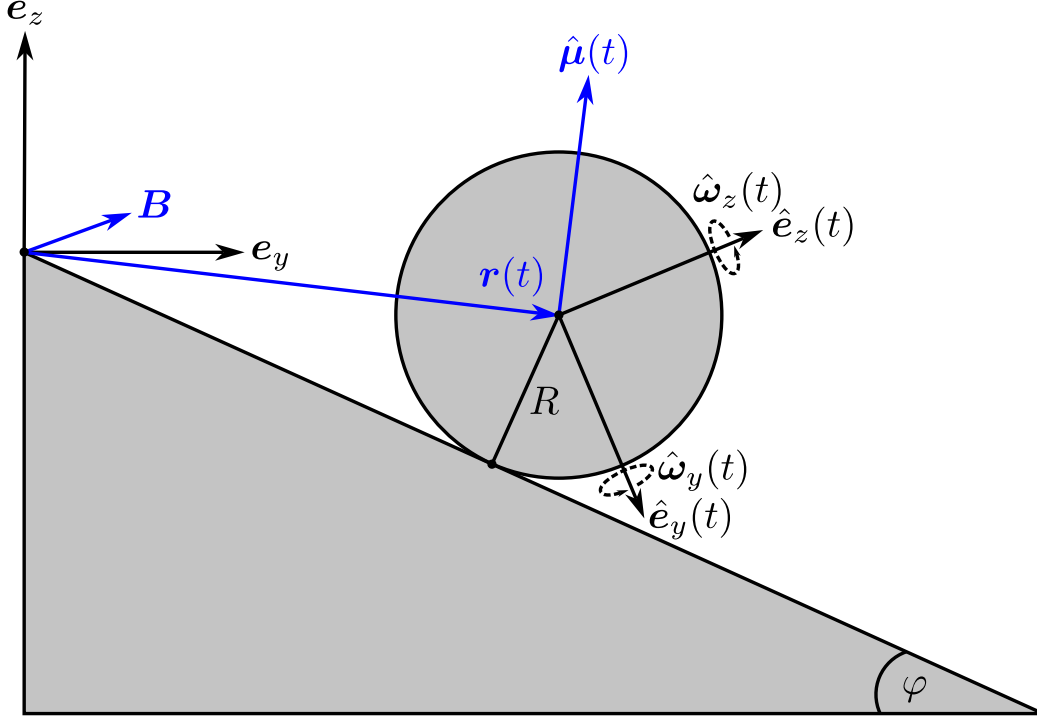


Figure 1: Depicted is the yz -projection of the incline system, the z -direction is orthogonal to the paper plane. The incline has an inclination angle φ and the sphere is of radius R . Two orthonormal coordinate frames are considered: The fixed frame $\{\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z\}$ and the body frame $\{\hat{\mathbf{e}}_x, \hat{\mathbf{e}}_y, \hat{\mathbf{e}}_z\}$. Vectors expressed in the body frame carry hats and are time-dependent, stationary vectors not. Blue vectors might have a non-zero z -component and are therefore only projections. The outer magnetic field is represented by $\mathbf{B}$, the magnetic dipole vector of the sphere by $\hat{\boldsymbol{\mu}}(t)$, and the centre of mass measured from the fixed frame by $\mathbf{r}(t)$. The in-plane angular velocities of the body frame are shown as $\hat{\omega}_x$ and $\hat{\omega}_y$.

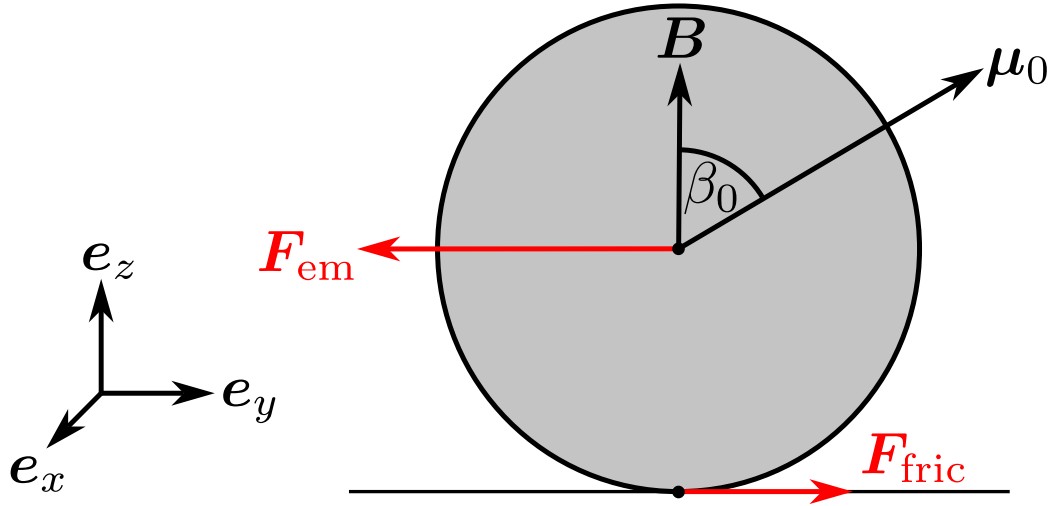


Figure 2: Shown is a sketch of the flat system with sufficiently small initial displacement angle β_0 between the outer magnetic field $\mathbf{B}$ and the initial magnetic dipole vector $\boldsymbol{\mu}_0$. Note that the system is effectively reduced to two dimensions. The red vectors indicate the oscillation-driving electromagnetic force $\mathbf{F}_{\text{em}}$ and the damping frictional force $\mathbf{F}_{\text{fric}}$. The frictional force includes aerodynamical drag and rolling resistance.

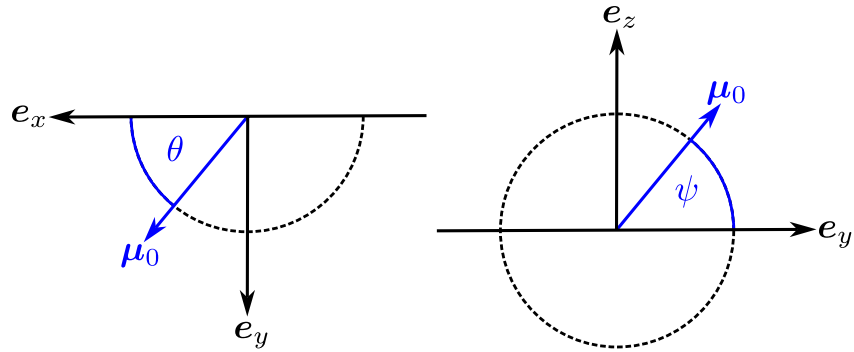


Figure 3: Parametrizations of $\boldsymbol{\mu}_0$ in F and G.

2 Figure Family A: Appendix

These figures are results of a proof-of-concept toy model: The incline without electromagnetic interaction. The goal is to compare numerical results to analytical equations.

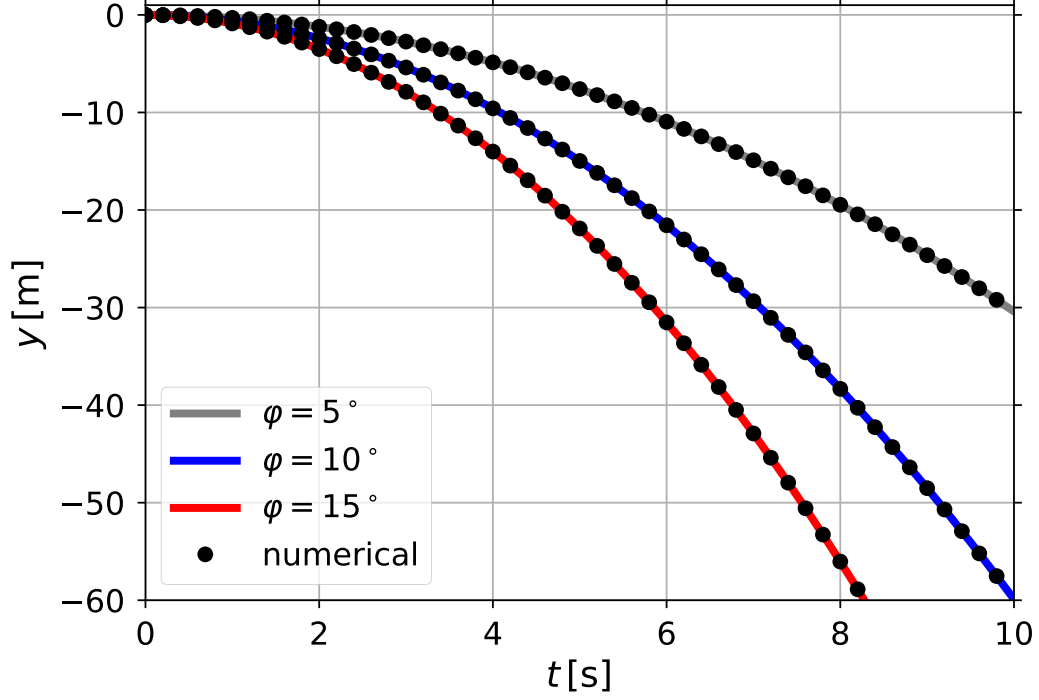


Figure A1: Shown are different evolutions of the y -projection of the centre-of-mass coordinate $\mathbf{r} = (x, y, z)^\top$ over time t for different incline angles φ without friction. Lines were calculated from the analytical solution and compared to the numerical results, represented as dots.

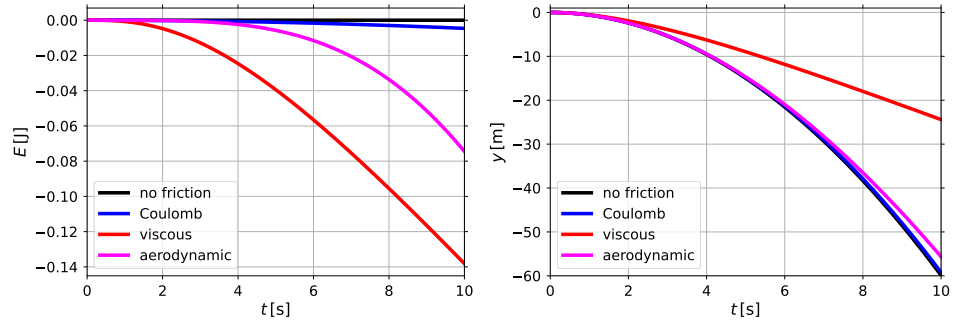


Figure A2/A3: The left plot shows the energy evolution $E(t)$ of the system over time t , while the right one depicts the y -COM-coordinate evolution. The data originates from the numerical simulation program. Compared are different friction models, where *Coulomb* represents dry rolling resistance, *viscous* represents a linear damping approximation to rolling resistance, and *aerodynamic* represents spherical aerodynamic drag.

3 Figure Family B

4 Figure Family C

5 Figure Family F: In Paper

The fixed non-standard parameters were:

- $\Delta t = 1s$
- $\phi = 10^\circ$
- Rolling Resistance on, Aerodynamical drag on

while the initial magnetic dipole moment was parametrized as

$$\boldsymbol{\mu}(t = 0) = \mu(\sin \varphi, \cos \varphi, 0)^\top$$

with polar angle φ and modulus $\mu = \|\boldsymbol{\mu}\|$. For each modulus $\mu \in \{0.0, 0.1, \dots, 2.5\}$, the polar angle was varied for $\varphi \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ in $\frac{\pi}{399}$ -steps, yielding $26 \cdot 400 = 10400$ data sets. The goal was to measure the influence of the magnetic moment on the deviation of the trajectory from a straight path.

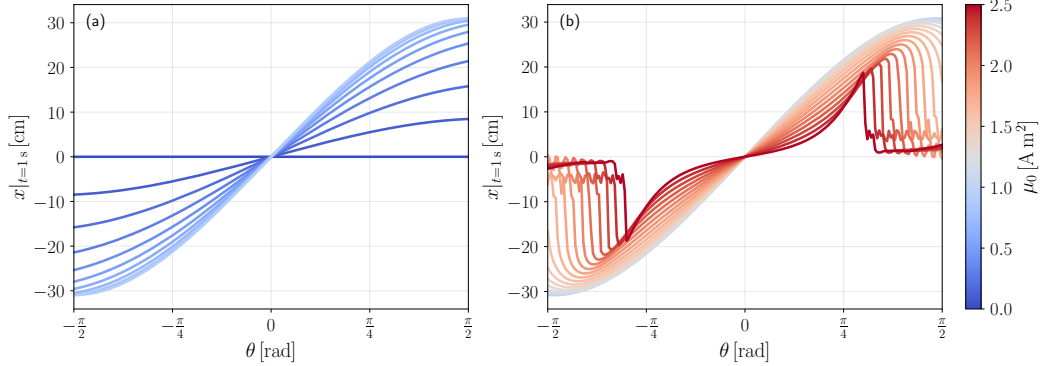


Figure F1: Final x -deviation w.r.t. polar angle and modulus. Figure (a) and (b) both show the x -value of a trajectory after $t = 1s$ depending on the initial polar angle. Each graph represents a modulus of $\boldsymbol{\mu}$, indicated by the color bar. The first ten graphs in (a) show that the trajectorial deviation increases non-linearly with the angular misalignment between $\boldsymbol{\mu}$ and the yz -plane. As expected, the deviation is antisymmetric about the origin. For increasing values of μ and φ , the deviation saturates. In (b), the curves for the remaining moduli μ can be seen. In this range, the magnetic forces start to dominate over the gravitational force for high polar deviations. The extrema of the curves correspond to the point where trajectories begin to loop around the origin instead of following the incline.

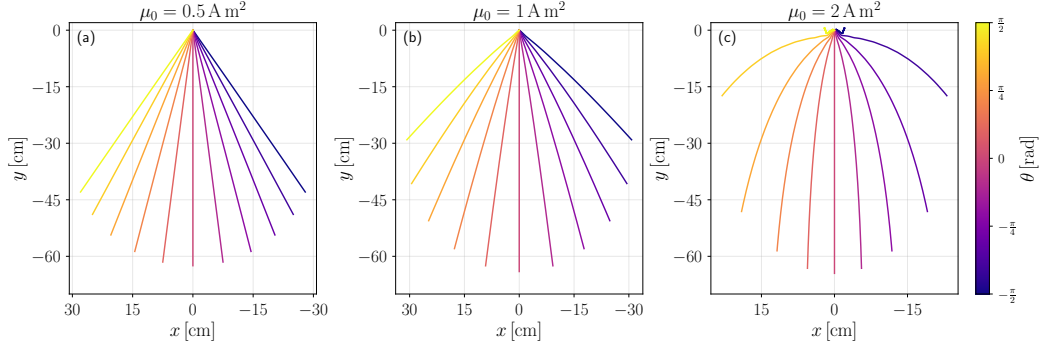


Figure F2: Chosen trajectories for fixed moduli. Figures (a), (b), and (c) show trajectories projected on the xy -plane for fixed values of μ and eleven equally spaced polar angle samples. It can be seen that in all cases, higher angular deviations correspond to stronger deflection of the sphere from a straight path. Increasing the modulus increases the curvature of the trajectories which evolve from almost linear paths to near-circular arcs. Figure (c) also shows the behaviour past the critical points of Figure F1(b) where the outmost paths form closed loops.

6 Figure Family G: In Paper

The fixed non-standard parameters were:

- $\Delta t = 1s$
- $\phi = 10^\circ$
- Rolling Resistance on, Aerodynamical drag on

while the initial magnetic dipole moment was parametrized as

$$\boldsymbol{\mu}(t=0) = (0, \sin \vartheta, \cos \vartheta)^\top$$

with polar angle ϑ and modulus $\|\boldsymbol{\mu}(t=0)\| = 1$. The goal was to analyze at which point the magnetic force, which aligns the magnetic moment with the outer field, is surpassed by gravity.

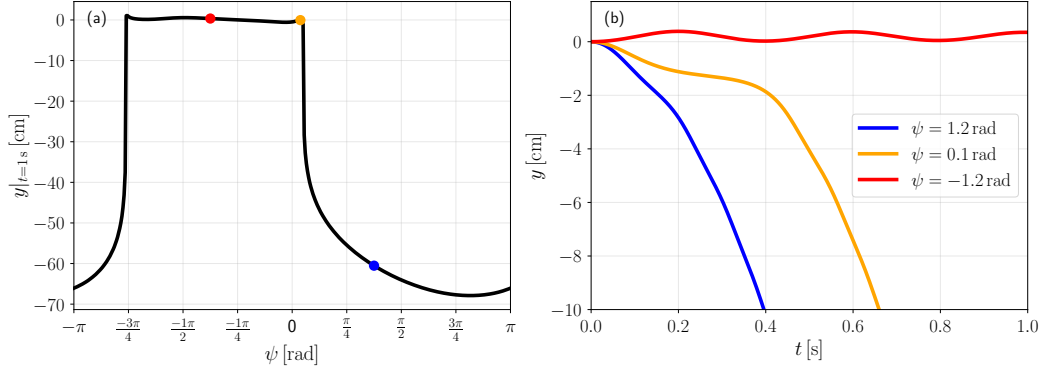


Figure G: Total y -displacement after $t = 1$ s for different polar angles (a) and associated representative y -projected trajectories (b). Figure (a) shows the total distance in y the sphere has rolled after one second depending on the polar misalignment of $\boldsymbol{\mu}$ and $\boldsymbol{B}$. There are three discernible regions: For almost all positive and a small range of negative deviations, the sphere covers distances characteristic for downward rolling motion. In the negative regime and around zero, the sphere is remaining around the origin. Between these regions, there are two steep boundary regions indicating an abrupt change between downward motion and oscillation around the origin. Figure (b) shows the evolution of the y -coordinate in time for three representative trajectories as indicated by the points in (a). The non-downward motion corresponds clearly to oscillation around the origin, while the downward motion shows only very weak signs of oscillatory motion in the beginning. The intermediate curve displays strong oscillatory motion in the beginning which is then overcome by gravity.

7 Figure Family H

These figures were generated with the following non-standard parameters:

- $\Delta t = 10s$
- $\phi = 5^\circ$
- Rolling Resistance on, Aerodynamical Drag on

The magnetic moment at $t = 0$ was again parametrized by

$$\boldsymbol{\mu} = \mu \cdot (\sin \varphi, \cos \varphi, 0)^\top.$$

The goal was to observe the behaviour of the magnetic moment vector in the rotating frame of the sphere. We considered two cases: Downward rolling motion for $\varphi = \frac{\pi}{5}$ and $\mu = 0.1Am^2$ and looping motion for $\varphi = \frac{\pi}{2}$ and $\mu = 1.5Am^2$. We lowered the slope and μ for the first one to obtain slower rotational velocities, making trajectories slightly easier to interpret. For the plots of $\boldsymbol{\mu}$ in time, we decomposed

$$\boldsymbol{\mu} = \boldsymbol{\mu}_{\parallel} + \mu_{\perp} \mathbf{e}_z.$$

The drawn out trajectories correspond to the $\mu_{\parallel}$ -component, i.e. the projection of $\boldsymbol{\mu}$ onto the xy -plane. The origin of the projected $\boldsymbol{\mu}(t)$ is the centre of mass of the sphere at time t . The trajectories of the COM were also plotted in arbitrary units, since they have to be scaled up to be visible in relation to the length of $\boldsymbol{\mu}$.

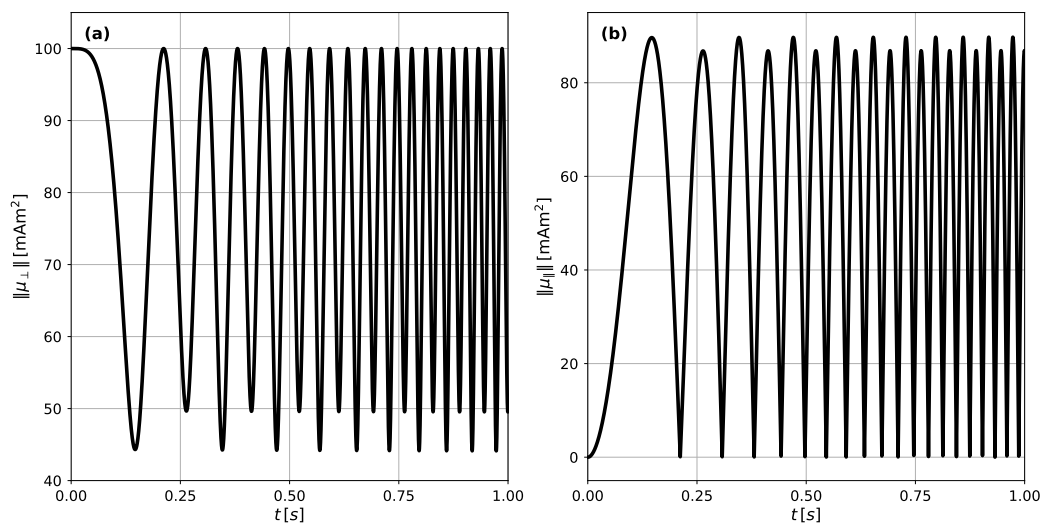


Figure H1:

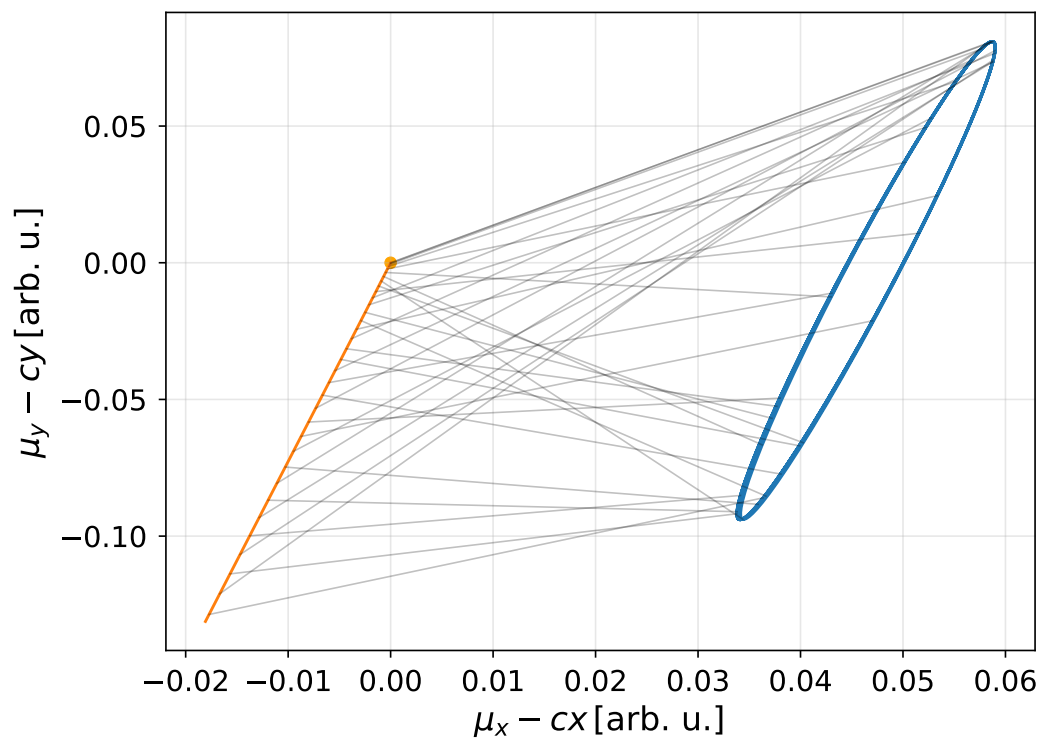


Figure H2:

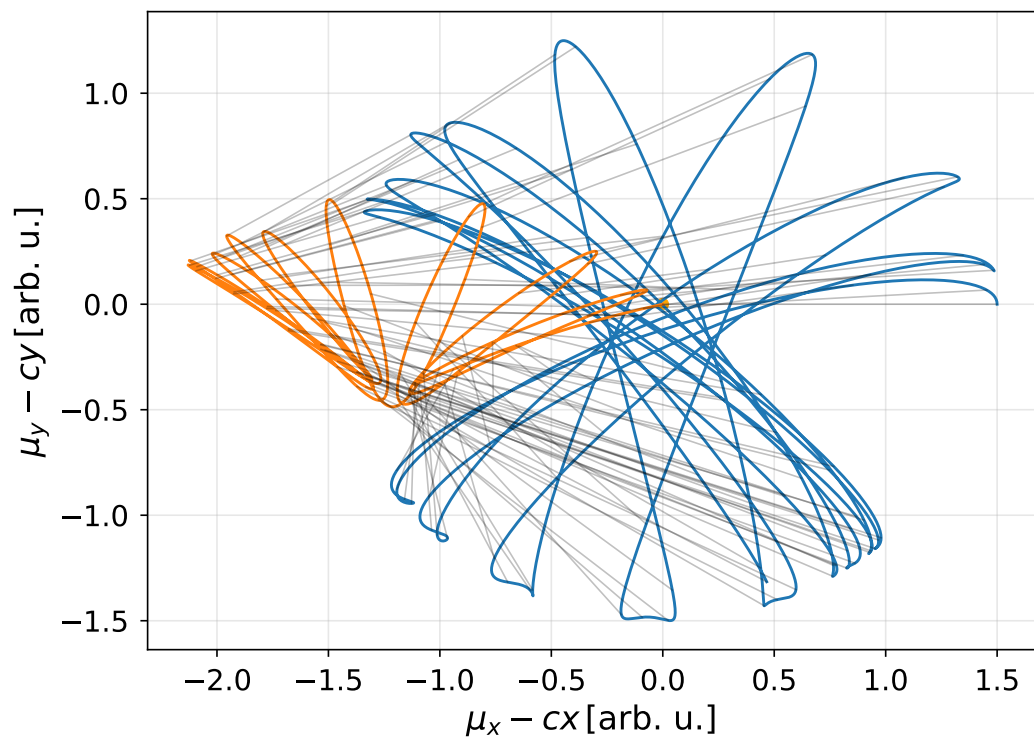


Figure H3:

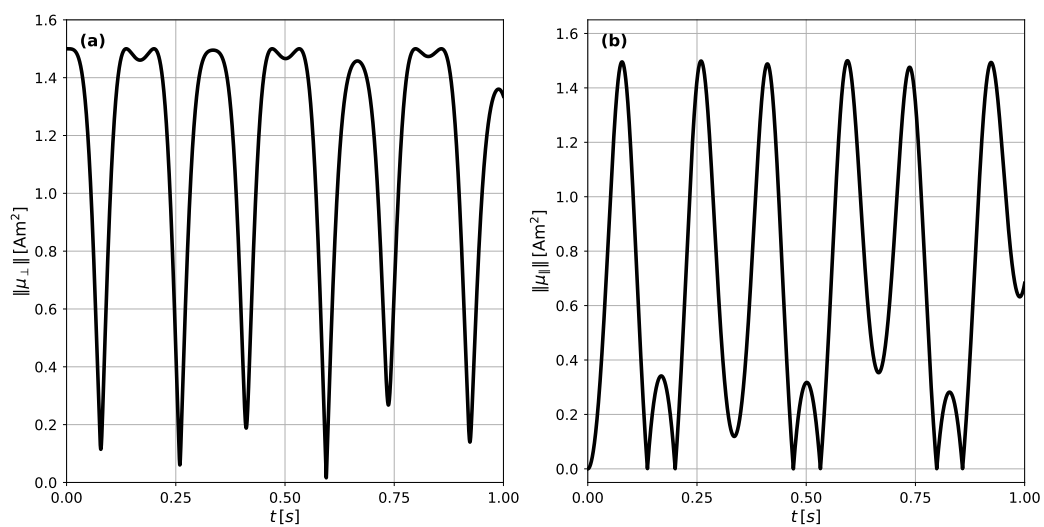


Figure H4:

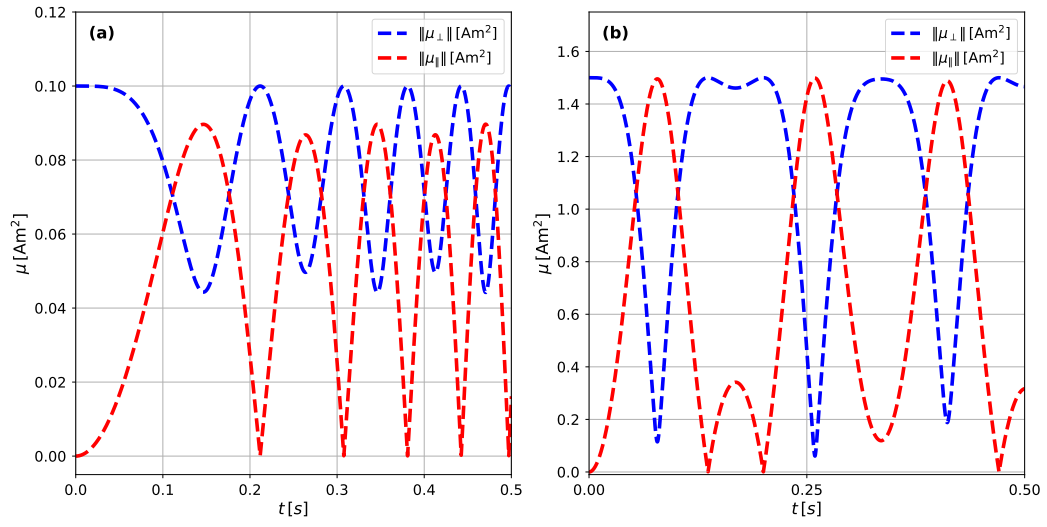


Figure H5:

8 Figure Family I: Not for paper

The non-standard parameters are

- $\Delta t = 10s$
- $\phi = 30^\circ$
- Rolling Resistance on, Aerodynamical Drag on

Goal of these calculations was to take a closer look at which inputs are close in orders of magnitude to the observed motion of the sphere. The idea is to assume that $\mathbf{B} = B_z \mathbf{e}_z$ and $\boldsymbol{\mu}(t=0)$ are almost perfectly aligned at rest, with a little polar perturbation in the xy -plane causing deflection of the path. To quantify the size of the perturbation, we considered $\varphi \in \{0, \frac{\pi}{20000}, \frac{\pi}{2000}, \frac{\pi}{200}, \frac{\pi}{20}, \frac{\pi}{2}\}$.

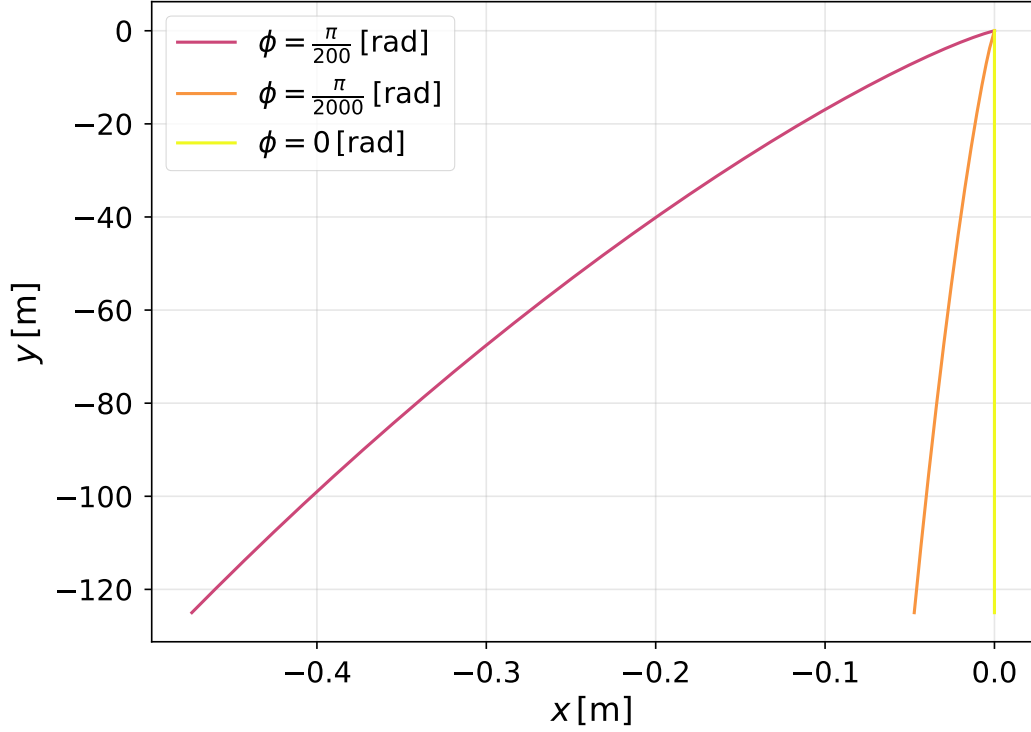


Figure I1: The paths show that only $\frac{\pi}{200}$ is in the correct range to be close to the experiment quantitatively.

9 Figure J

The non-standard parameters are

- $\Delta t = 1s$
- $\phi = 10^\circ$
- Rolling resistance on, AD on.

We compare the influence of the magnetic field with the influence of $\boldsymbol{\mu}$ by interchanging their roles from figure F and G. Set $\boldsymbol{\mu}_0 = (0, 0, 5 \cdot 10^{-5})^\top Am^2$ and

$$\boldsymbol{B}_\theta = B(\sin \theta, \cos \theta, 0)^\top$$

or

$$\boldsymbol{B}_\psi = B(0, \sin \psi, \cos \psi)^\top,$$

respectively. Then, we simply compare figure F1 and G with the new figures.

10 Figure K: In paper

The non-standard parameters are

- $\Delta t = 1s$
- $\phi = 20^\circ$
- RR on, AD on
- $\mu_0 = 1.0Am^2$
- No initial deviation: If the moment and field are not coplanar with the rolling direction, the rolling causes the misalignment after a few time steps automatically.

The goal is to investigate the general behaviour of the ball in roughly realistic settings. We used the NCEI model to derive the magnetic field vector at 53.5501721° north and 10.0013165° east (Hamburg, Centre) with a mean sea level elevation of $25m$ at the 12th of May, 2026. The total magnitude

was rounded to $5 \cdot 10^{-5}T$, close to the actual $50159.3nT$ of the NCEI. The magnetic field in the East-North-Up-Frame is then given by

$$\mathbf{B} = B \cdot (\cos I \sin D, \cos I \cos D, -\sin I),$$

where the NCEI predicts an inclination of $I \approx 68.6^\circ$ and a declination of $D \approx 4.2^\circ$.¹

11 Figure Family D-OLD: Replaced by F

The following figures are the results of running the simulation for $\Delta t = 1s$ simulation time with an incline of $\varphi = 10^\circ$ for all possible values of $\mu_x(t = 0)$ and $\mu_y(t = 0)$ in $[-1, 1]$ for increments of 0.1, meaning these are the results of 441 calculations.

¹The geographical spherical system is quite counterintuitive from the phycisists standpoint: The inclination is the dip of the field, so $I > 0$ means that the field points into the ground.

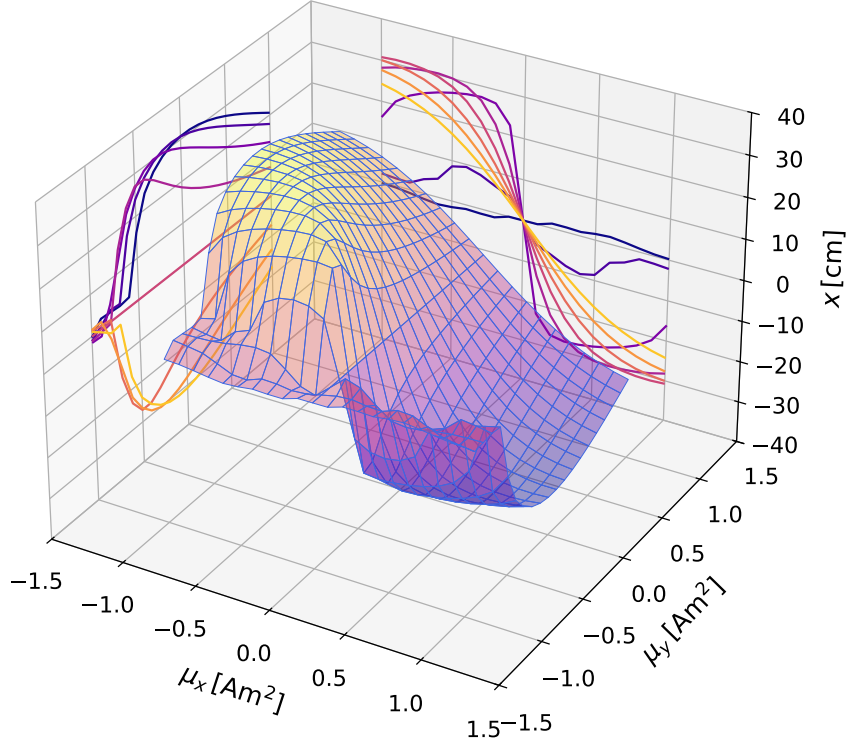


Figure D1: The surface represents the x -value the sphere attains after $t = 1$ s as a function of the initial x - and y -components of the magnetic dipole vector of the sphere. Since the x -direction is transverse to the rolling motion induced by gravity, it can be seen as a measure for the influence of the magnetic force on the trajectory. The graphs at the $\mu_y x$ -plane are slices of the surface for fixed values of μ_x , and respectively for the graphs on the $\mu_x x$ -plane with fixed μ_y .

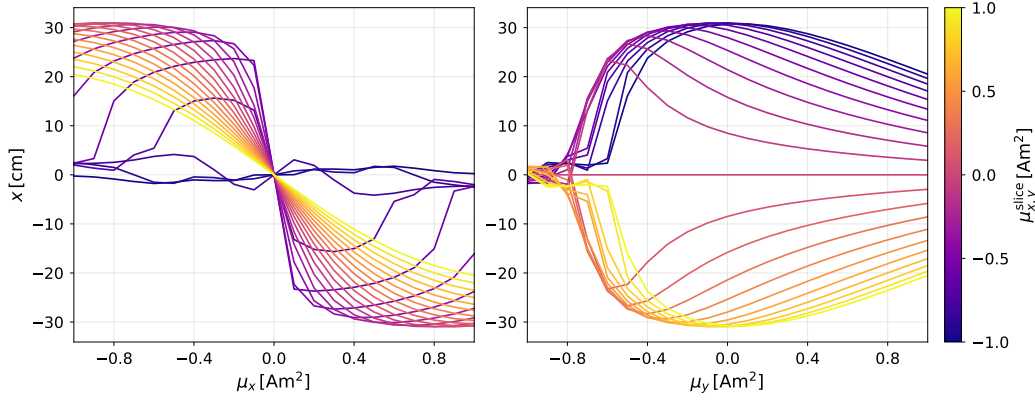


Figure D2: These plottings are enhanced versions of the slices of figure D1, left for different fixed μ_y and right for different fixed μ_x . Keeping μ_y fixed produces symmetrical deviations from the free rolling path depending on sign and magnitude of μ_x , as one would expect. A clear transition from regular sigmoid behaviour to non-smooth responses occurs at highly negative values of μ_y , at which the ball no longer rolls down, discussed more rigorously in D3. The slice graphs for fixed μ_x on the right show a characteristic constant curve for $\mu_x = 0$, as there can be no deviation from the purely gravitational path without a torque non-parallel to the yz -plane. All other slices display an asymmetry for positive and negative values. This is explained by the fact that for positive μ_y , the direction of rolling increases the misalignment of $\boldsymbol{\mu}$ with $\boldsymbol{B}$, slowing down the motion and hence also the deviation. On the other hand, negative μ_y -values cause $\boldsymbol{\mu}$ to align with $\boldsymbol{B}$, accelerating the motion in general and hence also the deviation in particular. A less regular asymmetry occurs for larger negative μ_y -values, which is again the critical region where the magnetic force dominates over and acts against gravity, causing looping paths instead of downward motion.

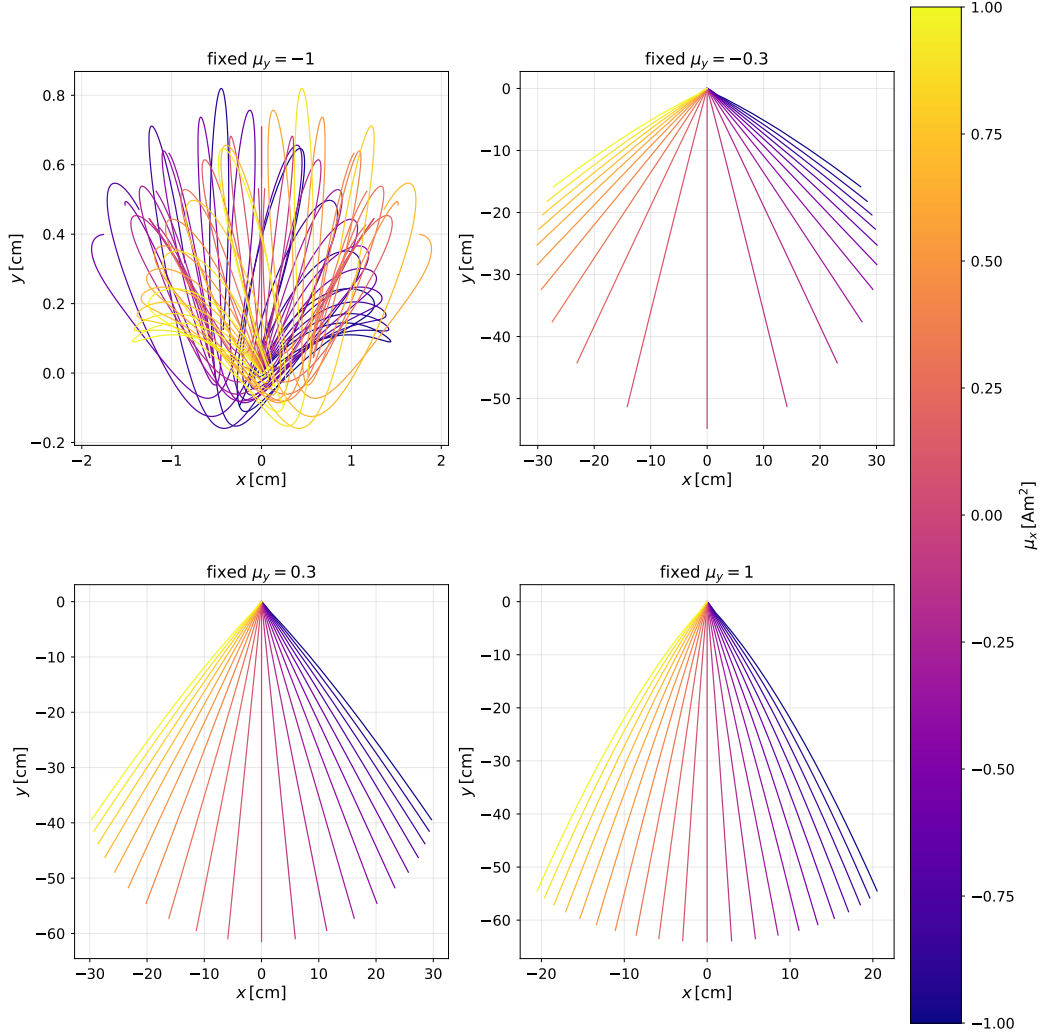


Figure D3: Visible are different trajectories of the sphere on the xy -plane for selected μ_y -slices and all μ_x -values corresponding to these slices. Here, the anomaly in D1 and D2 for large negative μ_y -values can be seen immediately. For smaller negative and all positive μ_y -slices, μ_y only controls the spread of the trajectories, while μ_x influences the deviation from a straight path. At $\mu_y = -1$, the magnetic force induced by the μ_y -component as soon as the μ_x -induced torque bends the trajectory out of the yz -plane overcompensates gravity and causes an upward motion. After that, the sphere moves in roughly elliptic loops.

12 Figure Family E-OLD: Replaced

The following figure shows the behaviour of the y -coordinate for different values of $\mu_z(t=0)$ in $[0, 1]$, incremented by 0.01-steps. The goal was to see the transition point at which the magnetic force overcomes the gravitational force, so $\mu_x(t=0) = \mu_y(t=0) = 0$.

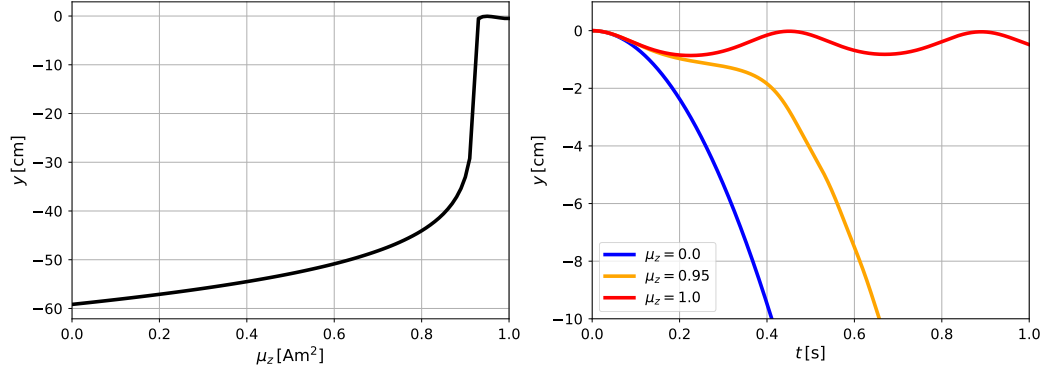


Figure E: The graph on the left shows the y -projection of the sphere's COM at $t = 1$ s as a function of initial μ_z -values. There are roughly three discernible regions: First, the y -value decreases, which is expected as the rolling motion increases the misalignment of $\boldsymbol{\mu}$ with $\boldsymbol{B}$. On the right, the y -trajectory for $\mu_z = 0$ is representative for this region. For increasingly high μ_z -values, a transition phase occurs. The representing graph at $\mu_z = 0.95$ shows that in this region, the sphere begins to oscillate slightly at first, but gravity soon overcomes the magnetic force. For $\mu_z \geq 0.96$, the magnetic force fully overcomes the gravitational force which only causes small oscillations around the origin, as seen by the representing graph for $\mu_z = 1$.